

FIREBREAK

****Catastrophes are too rare to learn from. Near-misses are the same process stopped early – so we learn the cascade from the failures that did not matter, to prevent the one that will.****

A decision system for interconnected city infrastructure. Given a stream of ordinary alarms, it answers the three questions no deployed system answers: which alarm becomes a catastrophe, what happens if we do nothing, and what is the cheapest action that stops it – with a deadline.

Code Merge Hackathon V2.0 · AI for Impact · Track: AI for Safety & Security → "The Chain Reaction"

The criterion

> ****An abnormality is dangerous exactly when its local branching ratio exceeds 1.****

>

> $n_i = \sum_j \alpha_{ij} \cdot s_j(t)$

Danger is not a property of an event. It is a property of its position in a network. A chiller running 3 °C hot is unremarkable in isolation and a catastrophe if it sits upstream of the room that runs the oxygen pumps.

α_{ij} – the expected number of failures at j caused by one failure at i – is estimated from ****mined near-misses****, not from expert-declared interdependency matrices and not from synthetic scenario sweeps. That is the contribution.

Where to start

You are	Read
****any agent, first****	[`EXECUTION.md`](EXECUTION.md) then [`docs/00-SHARED-CONTRACTS.md`](docs/00-SHARED-CONTRACTS.md)
a coding agent	[`AGENTS.md`](AGENTS.md), then your lane in [`lanes/`](lanes/)
wondering if it's true	[`docs/01-THESIS.md`](docs/01-THESIS.md) – including how to kill it
building the city	[`docs/02-CITY-SPEC.md`](docs/02-CITY-SPEC.md)
proving it works	[`docs/03-EVALUATION.md`](docs/03-EVALUATION.md)
presenting it	[`docs/04-DEMO-RUNBOOK.md`](docs/04-DEMO-RUNBOOK.md)
shipping it	[`docs/05-DEPLOYMENT.md`](docs/05-DEPLOYMENT.md)
worried	[`docs/06-RISKS.md`](docs/06-RISKS.md)

Lanes

Six, running in parallel from hour 6 behind a frozen contract.

Lane	Owns	The one thing
****A**** City & Simulator	``city/``, ``contracts/``, ``scenarios/``	Delays must emerge from physics, not from a constant
****B**** Inference	``inference/``	Learned $1/\beta$ must ****not**** correlate with ``nominal_delay_s``
****C**** Decision	``decision/``	Imports nothing from ``city/`` – enforced by AST test
****D**** Platform	``platform/``, CI, deploy	The redaction boundary; CI drives the six beats
****E**** Console	``console/``	The renderer computes nothing; 2D fallback from day one

| **F** Evaluation | `eval/` | Pre-register before results; report H4 even when it loses |

What we stand on

Declared loudly, never hidden.

- **InfraRisk** (`srijithbalakrishnan/dreaminsg-integrated-model`, BSD-3) – peer-reviewed water-power-transport interdependent failure simulation. Our substrate.
- **I³ / Icube** (Tsinghua FIB Lab, arXiv 2503.02890) – current SOTA for interdependent urban cascade prediction. **Our** baseline arm A5, not our strawman.
- **PI-GN-JODE** (arXiv 2603.20838) – physics-informed graph neural jump ODEs; source for the heavy-tailed cascade-size check.
- **Hawkes / ETAS** – the kernel, borrowed from seismology.
- **Aviation ASRS / NASA precursor work** – the licence for mining near-misses.

The gap all of them leave: **the literature predicts cascades; nobody picks the firebreak.**

The demo

Six keypresses, offline, nothing typed.

1. Chennai, five layers
2. Two alarms, **identical scores**, one flagged red
3. Do nothing → cascade to four hospitals, 60,000 people
4. Rewind, apply Firebreak's action → dies at hop two
5. Rewind, apply the human-obvious action → **still cascades**
6. The seven-arm table and the pre-registration diff

Beat 5 must fail. CI asserts that it does.

Status

Planning complete. No code yet. Open decisions in `EXECUTION.md` §12 – submission deadline, team size, and whether the Chennai topology is a real OSM extract or geo-anchored synthetic.